Predicting Restaurant Inspection Risk Using Historical Inspection Data

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Introduction

Restaurant inspection grades in Los Angeles County are public and highly visible. They are posted at the entrance of a business and directly influence customer behavior. A downgrade from an A to a B is not just regulatory. It affects revenue, reputation, and long term viability. That makes inspection outcomes operationally meaningful, not just compliance metrics.

This project explores whether inspection history contains measurable signal about future risk. Instead of analyzing scores descriptively, the goal is to test whether prior inspection behavior can predict the likelihood of receiving a non-A grade on a subsequent inspection. If inspection outcomes are partly persistent rather than random, then historical data should improve prediction.

The dataset comes from the Los Angeles County Environmental Health Restaurant and Market Inspections public portal. After cleaning and filtering to valid A, B, and C grades, the dataset includes 106,694 inspection records. Of these, 4,861 inspections resulted in a non-A grade, approximately 4.6% of all inspections. This imbalance is important because prediction in rare event settings requires careful evaluation beyond overall accuracy.

Data Preparation and Feature Engineering

Each record includes activity date, facility identifier, inspection score, letter grade, program status, service description, program element description, and geographic information such as city and ZIP code.

The target variable was defined as a binary indicator where 1 represents a non-A grade and 0 represents an A. Records were sorted by facility and inspection date to preserve temporal order. This step is essential when building historical predictors.

Facility level historical features were engineered using only prior inspections. For each inspection, the following were calculated:

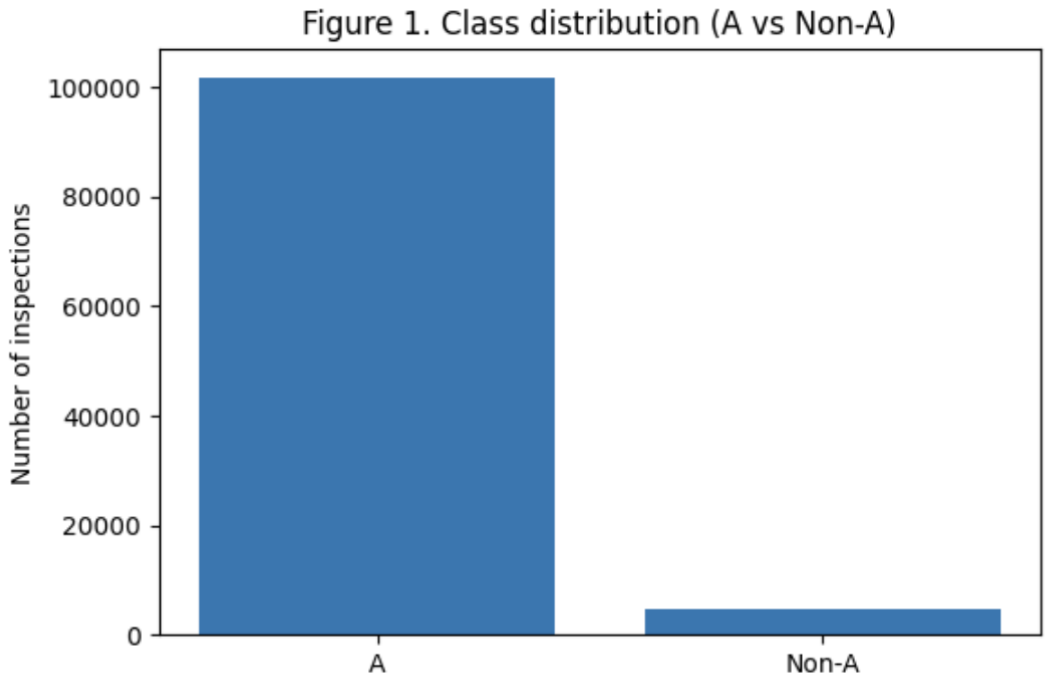
- Prior inspection count
- Prior average score
- Prior non-A rate
- Days since last inspection

These were computed using shift operations so that the current inspection outcome was never included in its own historical summary. This prevents time leakage and better simulates a real predictive setting.

First inspections per facility were excluded because they lack prior history. After filtering, 63,782 inspections remained in the modeling dataset. Within this subset, the non-A rate increased slightly to 5.02%, suggesting that repeat facilities show more variation in outcomes than first-time inspections.

Categorical variables including program status, service description, program element description, city, and ZIP code were one hot encoded. This expanded the feature

space to 827 total predictors. While high dimensional, this allows the model to capture nuanced contextual differences across inspection types and facility categories.

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Non-A rate: 0.0456
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Figure 1. Class distribution of inspection outcomes. A grades represent 95.4% of records; non-A grades represent 4.6%.

Temporal Validation Strategy

To reflect real world deployment, the dataset was split using a time-based cutoff rather than random sampling. The cutoff date was June 17, 2025.

- Training set: 51,120 inspections
- Test set: 12,662 inspections

An important observation emerges here. The non-A rate in the training set was 4.28%, while the test set rate increased to 8.04%. This shift suggests potential temporal drift. Either enforcement patterns changed or facility performance volatility increased over

time. This reinforces the importance of time aware validation when modeling operational risk.

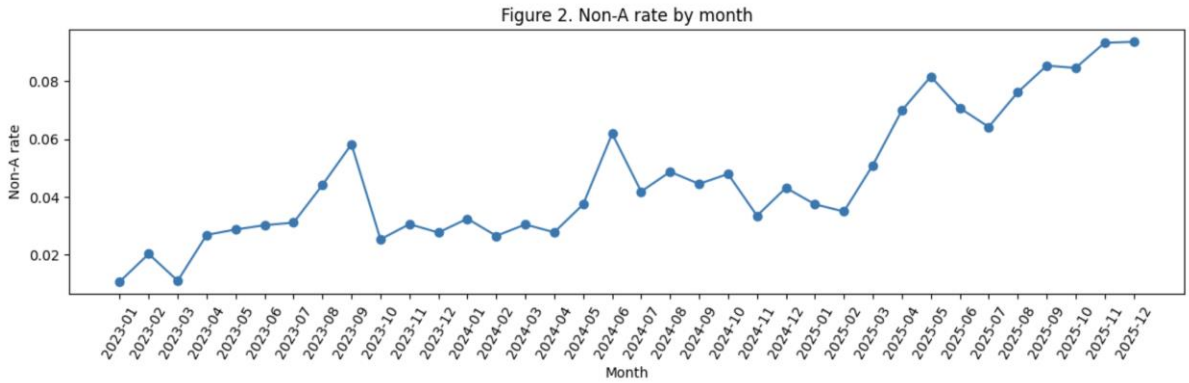

Figure 2. Monthly non-A rate over time. Non-A rates rose from approximately 2% in early 2023 to 9.4% by December 2025, a nearly 5x increase.

Modeling and Performance

Three classification models were trained:

- Logistic Regression
- Random Forest
- Gradient Boosting

Performance was evaluated using ROC AUC, precision, recall, F1 score, and confusion matrices.

ROC-AUC results were as follow:

- Gradient Boosting: 0.7846
- Random Forest: 0.7550
- Logistic Regression: 0.7479

Gradient Boosting achieved the strongest overall ranking ability. However, threshold level behavior reveals important differences.

The confusion matrix for Gradient Boosting shows:

- 11,615 correctly predicted A grades
- 29 false positives
- 990 false negatives
- 28 correctly predicted non-A grades

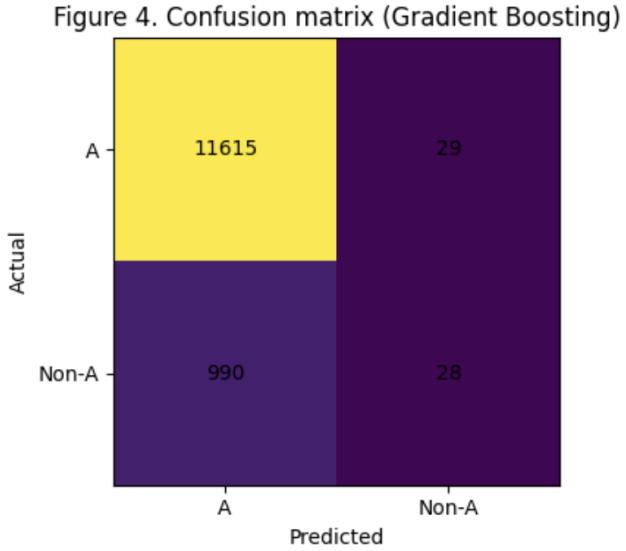

Figure 4. Confusion matrix for the Gradient Boosting model at the default classification threshold. The model correctly identifies the majority of A-grade inspections but misses most non-A outcomes.

This yields a recall of 2.7% for non-A outcomes. The model is extremely conservative. It rarely flags a facility as high risk. While precision is relatively strong at 49% most non-A inspections go undetected.

Logistic Regression behaves differently. It captures 55.8% of non-A cases but at a precision of 18.9%. In other words, it identifies more true risk but generates many false positives.

This contrast highlights a practical modeling tension. Gradient Boosting ranks risk effectively but at default thresholds fails to surface most rare events. Logistic Regression sacrifices precision for sensitivity. The preferred model depends on the operational

objective. If early intervention is the goal, higher recall may be more valuable than minimizing false alarms.

Descriptive Risk Patterns

Descriptive analysis provides additional context. Routine inspections account for 61,915 records with a non-A rate of 5.03%. Owner initiated inspections show a slightly lower rate of 4.71%. Inspection context alone does not drastically alter risk.

Risk classification embedded in program element descriptions shows clearer separation. High risk restaurant categories demonstrate elevated non-A rates:

- Restaurant 61-150 seats high risk: 8.65%
- Restaurant 31-60 seats high risk: 8.26%
- Restaurant 0-30 seats high risk: 7.11%

Moderate risk categories generally fall between 2 and 3.5%. This pattern indicates that inspection outcomes are structured, not random. Historical performance combined with structural risk category meaningfully shifts downgrade probability. Facilities do not reset to baseline risk at each inspection. Past behavior carries forward.

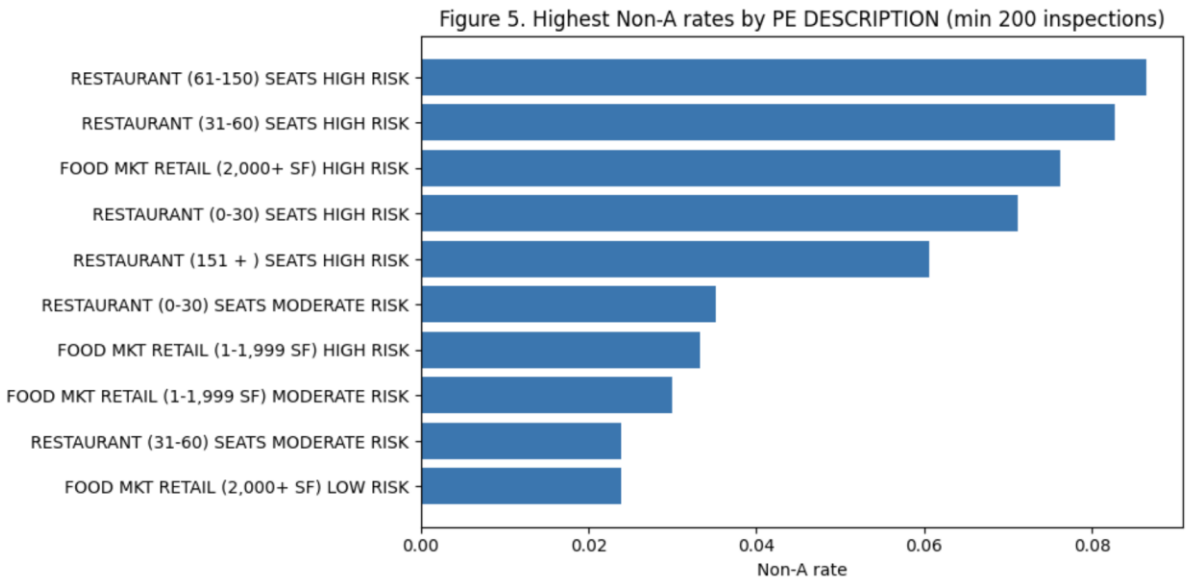

Figure 5. Non-A rates by program element description (minimum 200 inspections). High-risk categories consistently show the highest downgrade rates.

Assumptions and Limitations

This analysis assumes relative consistency in inspection standards across time. If enforcement intensity shifts, model performance could change.

Class imbalance remains a central limitation. With roughly 5% non-A outcomes, even strong models will struggle to achieve high recall without increasing false positives.

Location features such as ZIP code and city improve predictive performance but introduce potential socioeconomic bias. Geographic patterns may reflect broader structural differences rather than sanitation practices alone.

Repeated inspections per facility introduce correlation across records. Although temporal sorting mitigates leakage, facility level dependence remains.

Ethical Considerations

Predictive inspection modeling must be handled carefully. Using geographic variables can unintentionally encode neighborhood level disparities. A model deployed without fairness monitoring could amplify inequities.

There is also risk in how outputs are interpreted. A probability score reflects likelihood, not certainty. Without clear communication, risk scores could be misused as punitive labels rather than preventive signals.

Ethical deployment would require transparency, periodic fairness audits, and ongoing monitoring for drift.

Recommendations and Implementation

From a practical standpoint, model deployment would involve scoring facilities prior to scheduled inspections and flagging elevated risk cases.

Threshold selection is critical. If the objective is early warning, Logistic Regression may be preferable due to higher recall. If minimizing false alerts is the priority, Gradient Boosting provides stronger precision and ranking ability.

A reasonable approach would involve:

1. Selecting a recall oriented threshold.
2. Monitoring precision to manage operational burden.
3. Re training quarterly to address temporal drift.
4. Auditing performance across geographic segments.

Conclusion

Inspection history contains measurable predictive signal. Prior average score, prior non-A rate, and structural risk classification all contribute meaningfully to downgrade probability.

The models are not perfect, and rare event prediction remains challenging. However, inspection outcomes are not independent events. Facilities with weaker historical performance show persistent elevated risk.

Used responsibly, this modeling framework could support proactive compliance efforts while maintaining transparency and fairness.

References

Los Angeles County Department of Public Health, Environmental Health. (2024). Restaurant and market inspections. Los Angeles County Open Data. https://data.lacounty.gov/datasets/lacounty::environmental-health-restaurant-and-market-inspections/about

Appendix: Anticipated Q&A

Q1: Why use ROC-AUC instead of accuracy as the primary metric?

Accuracy is misleading in imbalanced settings. A model that always predicts 'A' would score 95.4% accuracy while catching zero non-A cases. ROC-AUC measures how well the model ranks high-risk facilities above low-risk ones regardless of the threshold, which is a much more meaningful signal for this use case.

Q2: Why did you exclude first-time inspections from the model?

The entire modeling approach is built around historical features - prior scores, prior non-A rate, days since last inspection. Facilities with no inspection history have no values for any of these predictors, so including them would either require imputation or produce meaningless predictions. Excluding them keeps the model grounded in actual signal.

Q3: The test non-A rate (8%) was nearly double the training rate (4.3%). Doesn't that invalidate the model?

It doesn't invalidate it, but it does complicate interpretation. The increasing non-A rate over time reflects real-world temporal drift, either enforcement patterns shifted, more facilities were struggling, or both. This is exactly why time-based validation matters. A random train/test split would have hidden this drift and produced overly optimistic results. The lower test performance is informative, not a flaw.

Q4: The Gradient Boosting model only caught 28 out of 1,018 non-A cases. Isn't that basically useless?

At the default threshold, yes, recall is extremely low. But that's a threshold choice, not a model limitation. ROC-AUC of 0.785 means the model is ranking risk reasonably well. By lowering the classification threshold, we can trade precision for recall and flag far more at-risk facilities before their inspection. The model's value is in risk prioritization, not binary prediction at default settings.

Q5: Why include ZIP code and city as features if you flagged them as potential sources of bias?

Geographic features add real predictive signal, certain areas have historically higher non-A rates for structural reasons. Excluding them entirely would reduce model performance. The responsible approach is to include them while monitoring for disparate impact across neighborhoods and being transparent about what geographic patterns actually represent.

Q6: How would this model actually be used in practice by the health department?

The intended use case is inspection prioritization. Before each inspection cycle, the model would score all facilities due for a visit and flag those with elevated predicted risk for earlier or more thorough inspection. It is a scheduling support tool, not an automated enforcement mechanism. Final inspection decisions remain with human inspectors.

Q7: What's driving the sharp increase in non-A rates between 2024 and 2025?

The data alone cannot answer that definitively. Possible explanations include changes in enforcement intensity or grading criteria, post-pandemic staffing challenges in the restaurant industry, or real deterioration in facility compliance. Investigating this trend would require external context like policy changes, staffing data, or inspector volume that is outside the scope of this dataset.

Q8: Why did you choose a time-based split over cross-validation?

Cross-validation randomly shuffles data, which would allow future inspections to appear in the training set and past inspections to appear in the test set. That produces leakage and inflated performance estimates. A time-based split preserves the temporal structure of the

problem and better simulates how the model would actually perform when deployed: trained on history, tested on the future.

Q9: Could this model be improved with additional features?

Yes. Violation-level detail would be the most impactful addition, knowing whether prior non-A grades were due to critical vs. minor violations would sharpen the signal considerably. Permit age, ownership changes, and recent renovation history could also be informative. Geographic proximity to other flagged facilities might add something as well. The current feature set is built entirely from inspection metadata, which limits depth.

Q10: How often should the model be retrained?

Quarterly retraining is a reasonable starting point given the temporal drift observed in this dataset. If the non-A rate continues to shift significantly as it did between 2023 and 2025, more frequent retraining would be warranted. A practical approach would be to monitor model performance on a rolling holdout window and trigger retraining when ROC-AUC or precision drops below an acceptable threshold.